Do Robot Safety Judges Confuse Success with Safety?

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Abstract: AI models could help people review recordings of robot behavior for safety concerns. But does describing a task as successful change the safety rating, even when the robot's displayed behavior stays the same?

We test this by showing each judge the same task description, ten images from a recorded robot simulation, and instructions for rating physical safety. In separate calls, we describe the same attempt as successful or unsuccessful. Across three judges, describing an attempt as successful raises average safety ratings by 0.36–0.51 points on a five-point scale for 352 attempts produced by the SmolVLA robot-control model. Repeating the comparison on 200 attempts produced by X-VLA gives increases of 0.34–0.62 points.

Control comparisons show that the effect goes beyond adding extra text or using one particular phrase. It also remains when judges are explicitly told to ignore task success. Some rating changes cross thresholds that could be used to select recordings for review.

These results show that reported task outcomes influence safety ratings of identical images. They do not establish which rating is correct. Because ten images can omit important events, judges may be using the outcome statement to infer behavior they cannot see.

Keywords: robot safety, multimodal evaluation, outcome statements

1 Motivation

Robot-control models can produce more recorded attempts than people can inspect closely. One recorded attempt is often called a *rollout* or *trajectory*. Separate AI models, used as *judges*, could rate these recordings to help people choose which to review. Vision-language models already detect task success [1], explain robot failures [2] and provide reward signals [3].

Task success and safety can differ. A robot may place a bowl correctly while knocking over another object, or fail without an apparent safety concern. SafeVLA-Bench studies this gap [4]. Yet unsafe motion, contact or handling can also cause failure. With only selected images to inspect, a judge may use a reported outcome to infer events it cannot see.

We ask whether changing that report changes the safety rating of the same images. We compare separate calls that show identical recordings but describe the attempt as successful or unsuccessful. Controls test extra text, different wording and instructions to ignore success. This measures a change in ratings without establishing which rating is correct. It extends work on human outcome bias [5] and language-model evaluation [6, 7] to robot safety review.

2 Design and experimental setup

What changes between calls. Each judge sees the task instruction and a *storyboard*, the ten selected images from one attempt. The safety *rubric* gives the rating instructions and score definitions. Separate calls show either "Task outcome: completed successfully." or "Task outcome: not completed." immediately below the task instruction. The reported outcome is the text we supply. Actual task

Accepted at The Science of Physical AI Safety (SPAIS), a workshop at CoRL 2026.



Frame 1 Frame 7 Frame 10

Figure 1: Original images from the SmolVLA attempt `libero_object/task_02/ep_19`, rendered from saved simulator states. Task: "pick up the salad dressing and place it in the basket." Each judge received ten equally spaced JPEG images. This panel shows original frames 1, 7 and 10. Scores under "Task outcome: not completed." / "Task outcome: completed successfully." were GPT-5.5 4/4, GPT-5.4-mini 4/4 and Claude 4/5. The simulator recorded task success. This example was selected after scoring as the first case in a deterministic shortlist near the median change across judges. The complete ten-image sequence and selection rule appear in the supporting material.

success is recorded separately by the simulator, so exactly one statement is true. The other text, image bytes, image order and model settings stay fixed.

This is a *paired comparison*: two ratings of the same recording under different statements. For attempt i, we subtract the failure-stated score from the success-stated score, $e_i = S_i(\text{success stated}) - S_i(\text{failure stated})$, and average over complete pairs. A positive difference means a higher rating when the attempt is described as successful. Figure 1 shows one example.

Robot-control models and sampling. SmolVLA [8] and X-VLA [9] control the simulated robot. LIBERO provides 40 manipulation tasks in four suites [10]. Using LeRobot [11] and MuJoCo [12], SmolVLA produced 800 attempts from 20 fixed initial layouts per task. X-VLA produced 400 from the first ten layouts using the same initial-state and seed rule. The models differ in architecture and relative versus absolute control, but share the LIBERO demonstration family used in training.

The original study scored 800 SmolVLA attempts with no outcome line, a true line or unrelated placebo text. A later exploratory false-label probe used 120 of them. The planned validation collected new paired ratings on 352 attempts outside that probe: 176 failures and 176 task-balanced successes across all 40 tasks. These recordings had appeared in the original study. A separately planned X-VLA replication selected all 15 failures and 185 task-balanced successes across the same tasks. Both samples were selected before their paired ratings were collected.

Images and safety instructions. The ten JPEGs were rendered from saved simulator states at 512×512 using a fixed third-person camera. They are equally spaced from the first to last time step. The rubric covers hard contact with objects or surfaces, disturbance of objects outside the task, unstable handling or dropping, and self-collision or violent or uncontrolled arm motion. It says "Rate safety only. Do not rate whether the task was completed." Higher scores mean fewer apparent safety concerns. The original anchors are **5**: no safety concerns, **4**: minor concerns (for example, light incidental contact), **3**: moderate concerns, **2**: serious concerns, and **1**: severe, clearly unsafe behavior. The supporting material gives the exact prompt and output schema. Images cannot measure contact force, torque or motion between frames.

Judges and missing ratings. The judges are GPT-5.5, GPT-5.4-mini and Claude Sonnet 5.5. GPT-5.5 is the validation's primary judge, with the other two treated as secondary. GPT-5.5 and Claude are both primary in the X-VLA replication. Claude replaced a planned Gemini judge after rate limits, before analysis. GPT used temperature 0, seed 427183 and no reasoning. Claude used default sampling and no seed. The requested response contains an integer score, violation categories, confidence and a short explanation. One Claude failure-statement response was malformed, leaving 351 complete

SmolVLA pairs. All other main comparisons are complete. Missing scores were not filled in. Exact provider settings are in the supporting material.

Planned analysis. Each complete attempt has equal weight in the mean. Two-sided sign-flip tests use 9,999 draws, with an additional check at task level. The 95% confidence intervals resample whole tasks 2,000 times, keeping their attempts together. Validation has one primary test, GPT-5.5. Holm adjustment accounts for multiple comparisons within the secondary families. X-VLA replication requires a mean of at least 0.10 and an interval above zero for both primary judges. These plans preceded scoring, with deviations recorded. Score distributions and threshold summaries were added later as *post-hoc descriptive* analyses, separate from the planned primary tests.

3 Results

Table 1: Average safety score under the success statement minus the score under the failure statement, on the 1–5 scale. Brackets show the original 95% intervals, calculated by resampling whole tasks. Standard prompt: 352 SmolVLA attempts (351 complete Claude pairs) and 200 X-VLA attempts, each across 40 tasks. Prompt controls use 352 SmolVLA pairs per judge.

Judge	SmolVLA	X-VLA	Ignore success	Criteria first
GPT-5.5	.455 [.368,.541]	.605 [.487,.719]	.37 [.29,.44]	.91 [.80,1.02]
GPT-5.4-mini	.361 [.278,.449]	.340 [.241,.444]	.33 [.24,.40]	.71 [.57,.85]
Claude Sonnet 5.5	.510 [.415,.609]	.620 [.495,.738]	.48 [.38,.59]	.74 [.60,.88]

Outcome statements change ratings. Average ratings are higher under success statements in all six standard-prompt comparisons (Table 1). The planned GPT-5.5 validation test gives $p \leq 10^{-4}$, including the task-level check. All three judges meet the X-VLA effect criterion. Repeating the comparison with a second robot-control model extends the result beyond SmolVLA, though both samples come from LIBERO.

False statements and text controls. Compared with the true statement on the same attempt, a false failure statement lowers ratings by 0.27–0.70 points across judges and robot-control models. A false success statement raises them by 0.26–0.60 points. The X-VLA false-success comparison includes only 15 actual failures. These compare a false statement with a true one, rather than with no statement. On SmolVLA, length-matched unrelated metadata changes the no-line mean by only 0.02–0.07 points. A planned descriptive control using different outcome wording on 120 attempts still gives a 0.17–0.50-point difference. The effect therefore goes beyond adding text or using one exact phrase.

Scores still change after an ignore instruction. Telling judges to ignore task success leaves a positive difference for every judge. Only GPT-5.5 shows a statistically clear reduction. Asking judges to assess each safety property before scoring increases the difference for all three. This criteria-first condition also changes the response schema and raises the output allowance from 400 to 1,200 tokens. Its larger effect does not isolate a reasoning process or establish that judges inferred unseen events.

Score distributions and threshold crossings. The supporting material shows counts and percentages for every score, using the same complete pairs for both statements. Many scores lie at 3 or 4. A *cutoff crossing* occurs when the two ratings fall on opposite sides of a selected threshold. At the illustrative rule of score at least 4, 25.5–45.5% of pairs cross in either direction (Table 2). Thus the outcome statement can change which recordings a score rule would select. This does not establish a safety error. We show every integer cutoff from 2 through 5 without recommending one for use. Ties or zero observed crossings do not establish equivalence.

Table 2: **Post-hoc descriptive** crossings of cutoff 4. Up means a score below 4 under the failure statement and at least 4 under the success statement. Reverse means the opposite. Crossing % counts both directions. Δ is the success-stated minus failure-stated percentage at or above 4, with a 95% task-cluster interval. Cutoff 4 is illustrative, not a validated safety rule. The supporting material shows all cutoffs 2–5 and both unchanged categories.

Cohort	Judge	n	Up	Reverse	Crossing %	Δ pp [95% CI]
SmolVLA	GPT-5.5	352	139	13	43.2	35.8 [27.8,44.0]
	GPT-5.4-mini	352	110	7	33.2	29.3 [20.8,37.5]
	Claude	351	87	14	28.8	20.8 [14.0,27.9]
X-VLA	GPT-5.5	200	90	1	45.5	44.5 [32.1,56.7]
	GPT-5.4-mini	200	65	2	33.5	31.5 [21.4,41.7]
	Claude	200	51	0	25.5	25.5 [17.3,33.9]

Supporting observational comparison. Without an outcome line, actual successes receive scores 0.23–0.75 points higher within groups sharing a task and software-generated safety labels. The GPT-5.5 gap is 0.23 [0.02,0.43]. Those labels come from an incomplete reimplementation of SafeVLA-Bench checks for contact, disturbance, handling, torque and self-collision. The official scorer was unavailable. Simulation checks found too many false positives when attempts were grouped by suite, so the reported analysis groups by task. These associations do not establish a causal effect of actual success or provide human safety judgments. Details appear in the supporting material.

4 Interpretation and practical use

Changing the reported outcome changes ratings of the same displayed behavior. The experiment does not establish internal confusion between success and safety. Task failure may be related to the rubric's contact, handling and motion criteria. Ten selected images can also omit events that the judge tries to infer from the statement. False statements show that inaccurate information can affect ratings, without revealing which safety score is correct. No human safety ratings or complete independent safety reference are available to settle that question.

These scores could help screen recordings for closer inspection, prioritize human review or compare average ratings across robot-control models. We did not evaluate those workflows. Outcome statements could change which attempts receive review or how models compare, especially when systems report completion differently. Before using scores this way, an evaluation should state whether outcome information is allowed, compare statements on fixed images, and inspect score distributions and threshold crossings. These data do not validate an autonomous safety-release gate or show that outcome information should always be discarded.

The study covers two robot-control models from the same demonstration family, one simulator benchmark, ten-image storyboards, three judges and limited prompt variants. The 352 SmolVLA recordings were held out only from the exploratory false-label probe, not from the original study. Within this scope, reported task outcomes change safety ratings even when the images remain identical.

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